WES 265 HW 1

1. Problems from Intuitive Probability Book

a) 4.19

R : rain

S : Sewer overflows

F : Flooded road

$P(S|R) = 0.5$; $P(F|S) = 0.3$

$P(R) = 0.2$; $P(F) = ?$

Notice that $F \subset S \subset R$ (i.e. event of flooded road is a subset of event of sewer overflowing, which itself is a subset of event of rain)

Therefore, $P(S|R) = \dfrac{P(S)}{P(R)}$ (1)

$P(F|S) = \dfrac{P(F)}{P(S)}$ (2)

$P(F) = P(F|S) \times P(S)$

$\Rightarrow P(F) = P(F|S) \; P(S|R) \; P(R)$

$\Rightarrow P(F) = P(F|S) \times P(S|R) \times P(R)$

(from (2))

$\Rightarrow P(F) = 0.3 \times 0.5 \times 0.2$

$= 0.03$

Probability that road will flood is 3%

b) 4.28

Only outcomes of the experiment are:

- success
- failure
- don't know

$P(\text{success}) = 0.5$

$P(\text{not success}) = 1 - 0.5 = 0.5$

This is akin to a coin toss (2 events with equal probability of 0.5).

$$P(3 \text{ successes out of } 5) = \frac{\binom{5}{3}}{2^5}$$

$$= \frac{10}{32} = \frac{5}{16}$$

This can also be arrived at using the binomial probability law

These trials are independent Bernoulli trials.

d) 5.6

$$P_X[k] = \alpha p^k \text{ for } k = 2, 3, 4, \ldots$$

For $P_X[k]$ to be a valid PMF,

$$\sum_{k=2}^{\infty} P_X[k] = 1 \;\&\; 0 \le P_X[k] \le 1 \quad \forall\, k$$

$$\Rightarrow \sum_{k=2}^{\infty} \alpha . p^k = 1 \Rightarrow \alpha\left(p^2 + p^3 + p^4 \ldots\right) = 1$$

Since $\alpha p^k > 0$ for every k, it follows that $\alpha > 0$. ($p^k > 0 \;\forall$ even k)

Also since $\alpha p^k \le 1$, it follows that $0 < p < 1$.

Further, $\sum_k p^k$ is a geometric progression sum, with the ratio, $p < 1$.

$$\therefore \alpha \sum^{\infty} p^k = \alpha . \frac{p^2}{\;} = 1$$

$$\therefore \alpha \sum_{k=2} \quad 1 - \rho$$

$$\Rightarrow \alpha = \frac{1-\rho}{\rho^2}$$

e) 5.20

$X \sim Pois(\lambda)$; $Y = 2X$

PMF of X : $P_X[k] = e^{-\lambda} \cdot \dfrac{\lambda^k}{k!}$, $k = 0, 1, 2 \ldots$

PMF of $Y = 2X$

By definition, the RV Y can only take even values.

$$P[X = k] = e^{-\lambda} \cdot \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, 3 \ldots$$

$$P[Y = 2X = y] = P[X = y/2]$$

$$= e^{-\lambda} \cdot \frac{\lambda^{y/2}}{(y/2)!}$$

$$\therefore P_Y[m] = \left\{ e^{-\lambda} \cdot \frac{\lambda^{m/2}}{(m/2)!}, \text{ for } m = 0, 2, 4 \ldots \right.$$

$$\left(\quad 0 \quad , \text{ otherwise} \right.$$

f) 5.31

Let X be the number of calls in a 1 minute interval.

λ = Arrival rate $\times$ T = 1×60

$$P(X > 100) = 1 - P(X \leq 100)$$

$$= 1 - \sum_{k=0}^{100} e^{-\lambda} \cdot \frac{\lambda^{-k}}{k!}$$

$$\Rightarrow 1 - e^{-\lambda}\left(1 + \frac{\lambda^{-1}}{1!} + \frac{\lambda^{-2}}{2!} + \frac{\lambda^{-3}}{3!} \cdots \frac{\lambda^{-100}}{100!}\right)$$

Numerically, $\lambda = 60$ & $\frac{\lambda^{-2}}{2!} \sim 10^{-4}$

$\frac{\lambda^{-3}}{3!} \sim 10^{-6}$. The higher power terms are numerically insignificant.

$$\Rightarrow 1 - e^{-60}\left(1 + \frac{1}{60}\right) \simeq 1$$

$\therefore$ It is almost certain that there will be more than 100 calls in a 1 minute interval

g) 11.10

$$X = A + U$$

$$U \sim \mathcal{U}(-1/2, 1/2)$$

$$SNR = \frac{E^2[x]}{Var(x)}$$

$$E[x] = E[A + U]$$

$$= E[A] + E[U]$$

$$= A + 0$$

$$Var(x) = Var(A + U)$$

Since A is not random,

$$Var(x) = Var(U)$$

$$= \int_{-1/2}^{1/2} x^2 \cdot 1 \cdot dx$$

$$= \left[\frac{x^3}{3}\right]_{-1/2}^{1/2} = \frac{1/8}{3} - \left(\frac{-1/8}{3}\right)$$

$$= \frac{1}{24} + \frac{1}{24} = \frac{1}{12}$$

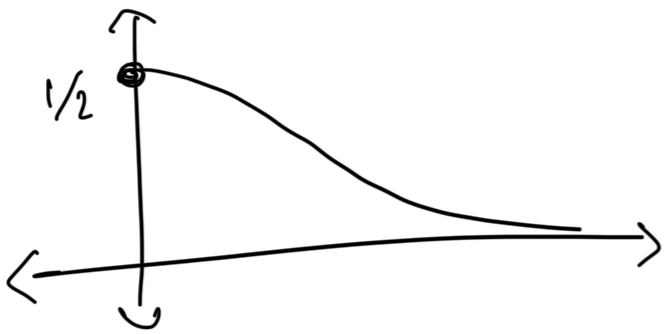
$$\therefore \quad SNR = 1000 = \frac{A^2}{1/12}$$

$$\Rightarrow \quad A = \sqrt{\frac{1000}{12}} = 10\sqrt{\frac{5}{6}}$$

$$\sim 9.1287$$

h) 11.27

$$P_X(x) = \frac{1}{2}\delta(x) + \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}} u(x)$$



The mixed PDF has 2 components:

- Discrete at $x = 0$ with the Dirac delta function giving it a probability of 0.5.
- Continuous for $x > 0$ with the random variable having a normal/Gaussian distribution.

$$E[X^2] = \int^{\infty} x^2 \cdot P_X(x) \cdot dx$$

$$-\infty$$

$$P_X(x) = 0 \quad \forall \; x < 0$$

$$\therefore \; E[x^2] = \int_0^\infty x^2 \cdot \frac{1}{2} \delta(x) \cdot dx \; +$$

$$\int_0^\infty x^2 \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \cdot dx$$

$$\Rightarrow \; E[x^2] = \frac{1}{2} \int_0^\infty x^2 \cdot \delta(x) \cdot dx \; +$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot \int_0^\infty x^2 \cdot e^{\frac{-x^2}{2\sigma^2}} \cdot dx$$

$$\int f(x)\delta(x) \cdot dx = f(0)$$

$$\Rightarrow \; E[x^2] = 0 + \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \int_0^\infty x^2 \cdot e^{\frac{-x^2}{2\sigma^2}} \cdot dx$$

$$-2$$

$$\Rightarrow E[x^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_0^\infty x \cdot x \cdot e^{\frac{-x}{2\sigma^2}} \cdot dx$$

Using uv rule :

$$\int u \cdot dv = uv - \int v \cdot du$$

$$u = x \; ; \; du = dx$$

$$dv = x \cdot e^{\frac{-x^2}{2\sigma^2}} \cdot dx$$

$$v = \int x \cdot e^{-\frac{x^2}{2\sigma^2}} \cdot dx$$

$$\frac{d}{dx}\left(e^{-\frac{x^2}{2\sigma^2}}\right) = e^{-\frac{x^2}{2\sigma^2}} \cdot \left(-\frac{\cancel{2}x}{\cancel{2}\sigma^2}\right)$$

$$= \frac{-1}{\sigma^2} \cdot x \cdot e^{\frac{-x^2}{2\sigma^2}}$$

$$\therefore \; V = \int x \cdot e^{\frac{-x^2}{2\sigma^2}} \cdot dx = -\sigma^2 \cdot e^{\frac{-x^2}{2\sigma^2}}$$

$$\Rightarrow \int u \cdot dv = uv - \int v \cdot du$$

$$= \left[-x\sigma^2 \cdot e^{\frac{-x^2}{2\sigma^2}} \right]_0^\infty -$$

$$\int_0^\infty -\sigma^2 \cdot e^{\frac{-x^2}{2\sigma^2}} \cdot dx$$

$$\Rightarrow E[X^2] = \left(0 + \sigma^2 \int_0^\infty e^{\frac{-x^2}{2\sigma^2}} \cdot dx \right) \frac{1}{\sqrt{2\pi\sigma^2}}$$

$$\Rightarrow E[X^2] = \frac{\sigma}{\sqrt{2\pi}} \int_0^\infty e^{\frac{-x^2}{2\sigma^2}} \cdot dx$$

$$\frac{x}{\sqrt{2}\sigma} \; ; \quad z = \frac{x}{2\sigma^2}$$

$$dz = \frac{dx}{\sqrt{2}\sigma}$$

$$\Rightarrow E[x^2] = \frac{\sigma}{\sqrt{2\pi}} \int_0^\infty e^{-z^2} \cdot \sqrt{2}\sigma \cdot dz$$

$$= \frac{\sigma^2}{\sqrt{\pi}} \int_0^\infty e^{-z^2} \cdot dz$$

Now. $\int_{-\infty}^{\infty} e^{-x^2} \cdot dx = \underline{I}$

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} \cdot dx \times \int_{-\infty}^{\infty} e^{-y^2} \cdot dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} \cdot dx \cdot dy$$

Converting to polar co-ordinates,

$x = r.\cos\theta$; $y = r.\sin\theta$

$r^2 = x^2 + y^2$

$\theta : 0 \to 2\pi$ & $r : 0 \to \infty$

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} . r . dr . d\theta$$

$$(dx . dy = r . dr . d\theta)$$

$$I^2 = \int_0^{2\pi} \left(\int_0^{\infty} r . e^{-r^2} dr \right) d\theta$$

$$= \int_0^{2\pi} \left[\frac{-e^{-r^2}}{2} \right]_0^{\infty} . d\theta$$

2π $\left[\quad \right]^{2\pi}$

$$= \int_0 + \frac{1}{2} \cdot d\theta = \frac{1}{2} \left[\theta \right]_0$$

$$I^2 = \pi$$

$$\therefore I = \int_{-\infty}^{\infty} e^{-x^2} \cdot dx = \sqrt{\pi}, \text{ and}$$

$$\int_0^{\infty} e^{-x^2} \cdot dx = \frac{\sqrt{\pi}}{2}$$ since e^{-x^2} is even & symmetric.

$$\therefore E[x^2] = \frac{\sigma^2}{\cancel{\sqrt{\pi}}} \times \frac{\cancel{\sqrt{\pi}}}{2}$$

$$\Rightarrow \frac{\sigma^2}{2}$$

This result makes sense because the mixed PDF is essentially the

normal distribution with mean $= 0$ & variance σ^2, cut in half for $x > 0$.

This PDF can be physically interpreted as the average power output of a half-wave rectifier with input being a noise sample with PDF $\mathcal{N}(0, \sigma^2)$.